



PRIMARY TREATMENT

of Tannery Effluent



Wastewater and Solid Waste
Management

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Why Effluent Is Hard to Treat

- 🧪 **High organic load:** Strong BOD/COD from beamhouse proteins.
- ☠️ **Sulphide:** Releases toxic H_2S gas if mixed with acid.
- ⚠️ **Chromium:** Toxic heavy metal; must be segregated & recovered.
- 🏠 **High salinity:** Chlorides/sulphates that biology cannot remove.
- ↔️ **Variable flow/pH:** Batch dumps swing pH 2.5 to 12 all day.

The biological stage is sensitive—toxins and shock loads must be removed FIRST.



Objectives of Primary Treatment



1. Remove Coarse Material

Protect downstream pumps, pipes, and sewers from severe clogging and mechanical damage.



2. Mix & Balance

Blend highly variable streams to produce a single, uniform, and predictable feed flow.



3. Adjust pH & Detox

Neutralise pH extremes and fully oxidise toxic sulphide to dampen shock loads.



4. Cut Organic Load

Significantly lower the BOD and COD load, reducing the cost and size of secondary treatment.

Collection: Three Tannery Streams

Stream	Main Source	Dominant Pollutants
Beamhouse Largest volume; strongest organic load	Soaking, liming, deliming, bating	Sulphide, lime, high pH, BOD/COD, suspended solids, protein.
Tanyard Most hazardous	Pickling, chrome tanning	Chromium, chloride, low pH, sulphate.
Post-tanning Moderate strength	Retanning, dyeing, fatliquoring, finishing	Syntans, dyes, fats, residual low-level chrome.

Critical Practice: Each stream is collected through its own separate drainage network so it can be treated accurately before mixing.

Keep the Streams Apart

Why it matters: Below ~pH 8, sulphide turns into dissolved H_2S ; below pH 6 it escapes as lethal gas. Lime liquors must be completely oxidised *before* encountering any acidic pH drops.

Chrome Drain

- Carries spent, acidic chrome tanning liquor.
- Sent directly to a dedicated **chrome recovery** unit.
- Chromium is precipitated, redissolved, and reused in the process.
- Critically keeps toxic heavy metals out of the main mixed effluent sludge.

General / Lime Drain

- Carries highly alkaline, sulphide-rich liquors from beamhouse operations.
- Sulphide must be **catalytically oxidised** as the very first chemical step.
- Strict separation ensures acidic streams never prematurely mix and trigger H_2S gas release.

Primary Treatment Sequence

- 1 Coarse (bar) screen:**
Removes large debris (hide scraps, plastics).
- 2 Grease & grit trap:**
Skims floating fats and settles heavy sand/grit.
- 3 Pumping station:**
Lifts the effluent to allow subsequent gravity flow.
- 4 Fine screen:**
Captures ~1 mm fibres, hair, and finer particles.
- 5 Equalisation & oxidation:**
Balances flow/load, adjusts pH, and destroys sulphide.
- 6 Coagulation–flocculation:**
Chemical addition to aggregate colloids, chrome & colour.
- 7 Primary clarifier:**
Settles the chemical flocs into sludge.
- 8 Sludge dewatering:**
Mechanically handles and concentrates settled solids.

Screening: First Mechanical Step

- ≡ **Coarse / bar screens:** Parallel bars with ~20–50 mm spacing. Uses self-cleaning mechanical rakes.
- ▣ **Fine screens:** ~1 mm aperture. crucial for removing finer hairs and hide fibres after pumping.
- 🔹 **Grease & sand trap:** Skims fats and settles grit before fine screening, protecting delicate downstream infrastructure.

Efficiency Note

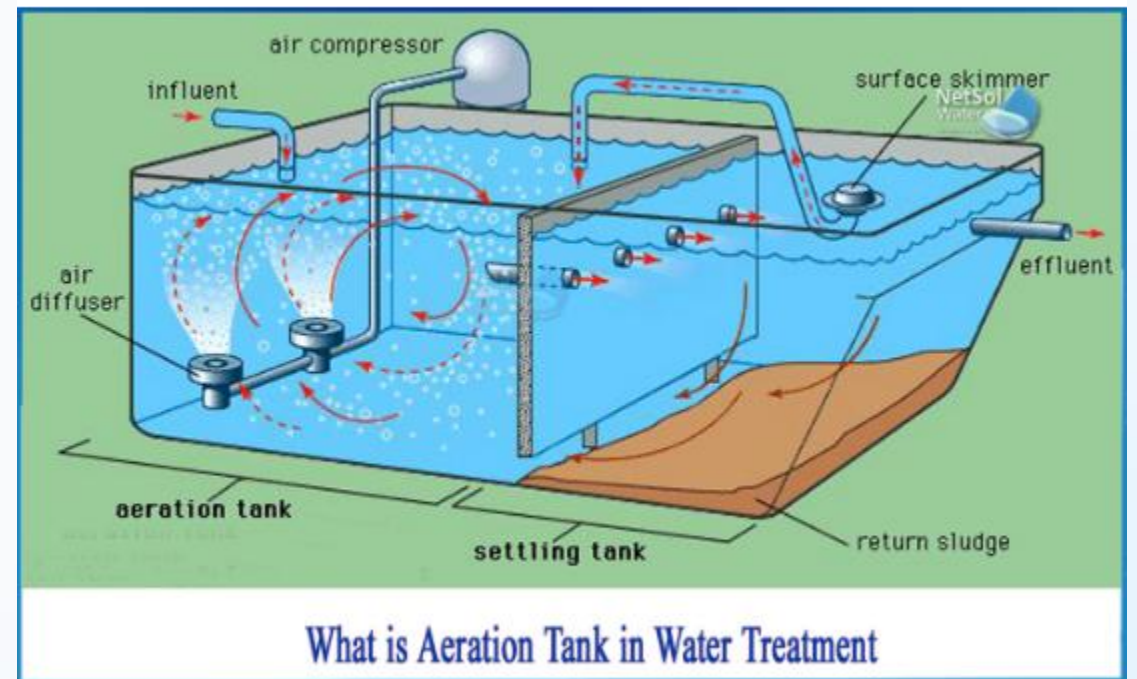
Screening alone removes **30–40%** of the gross suspended solids before complex treatment even begins.



Equalisation: The Heart of ETP

Tannery flow, pH, and pollutant levels swing sharply all day. The equalisation tank serves as an enormous buffer to smooth these into a steady, reliable feed.

- 🌊 **Flow balancing:** Constant output to clarifiers.
- ⚖️ **Load homogenisation:** Evens out BOD/COD spikes.
- 🔧 **Neutralisation:** Blends pH 11–12 and 2.5–4 to ~pH 8.
- 🔄 **Oxidation:** Diffused air system drives sulphide destruction.



Equalisation: Design & Mixing

Design Criteria & Mixing

Sizing rule: Tank volume should approximately equal the tannery's total daily discharge (keeps a full day's surge in reserve). Add ~20% for freeboard.

Continuous Mixing Methods:

- **Diffused air** (Highly preferred) — simultaneously mixes and oxidises sulphide.
- Venturi / jet ejectors.
- Mechanical stirring (Surface aerators are avoided due to severe odour release).

Worked Example

Given Flow = $200 \text{ m}^3/\text{day}$

Working Vol $\approx 200 \text{ m}^3$

+ 20% Freeboard $\rightarrow 240 \text{ m}^3$ total

Assume tank depth = 4 meters

Required Area = $240 \text{ m}^3 / 4 \text{ m}$

= 60 m^2

⇒ **Tank dimension $\approx 7.7 \text{ m} \times 7.8 \text{ m}$**

pH Correction & Neutralisation

Alkaline Streams

Primary Source: Lime and general beamhouse operations.

Typical pH Range: 11 – 12

↓ Requires dosing with acid (H_2SO_4) or captured CO_2 gas to safely lower the pH.

Acidic Streams

Primary Source: Pickle and chrome tanning operations.

Typical pH Range: 2.5 – 4.0

↑ Requires dosing with lime (CaO) or sodium hydroxide to raise the pH.

Target Neutralisation: pH 7.0 – 8.5

This is the optimal range for downstream biological treatment and effective metal precipitation.

Sulphide Oxidation

Chemical Reaction

Sulphide is converted into harmless sulphate using oxygen.



Catalytic air oxidation: Employs a manganese-salt catalyst with intense air blown through a jetox venturi system.

Process Parameters

-  Requires a batch treatment time of about **8 hours** to complete.
-  Effectively achieves a residual sulphide level of **< 1 mg/L**.
-  **Intensive oxygen demand:** Needs ≈ 2 grams of O_2 per 1 gram of sulphide.
-  Preferred over alternative methods: Ferric salts produce excessive sludge, and H_2O_2 is financially prohibitive.

Coagulation & Flocculation

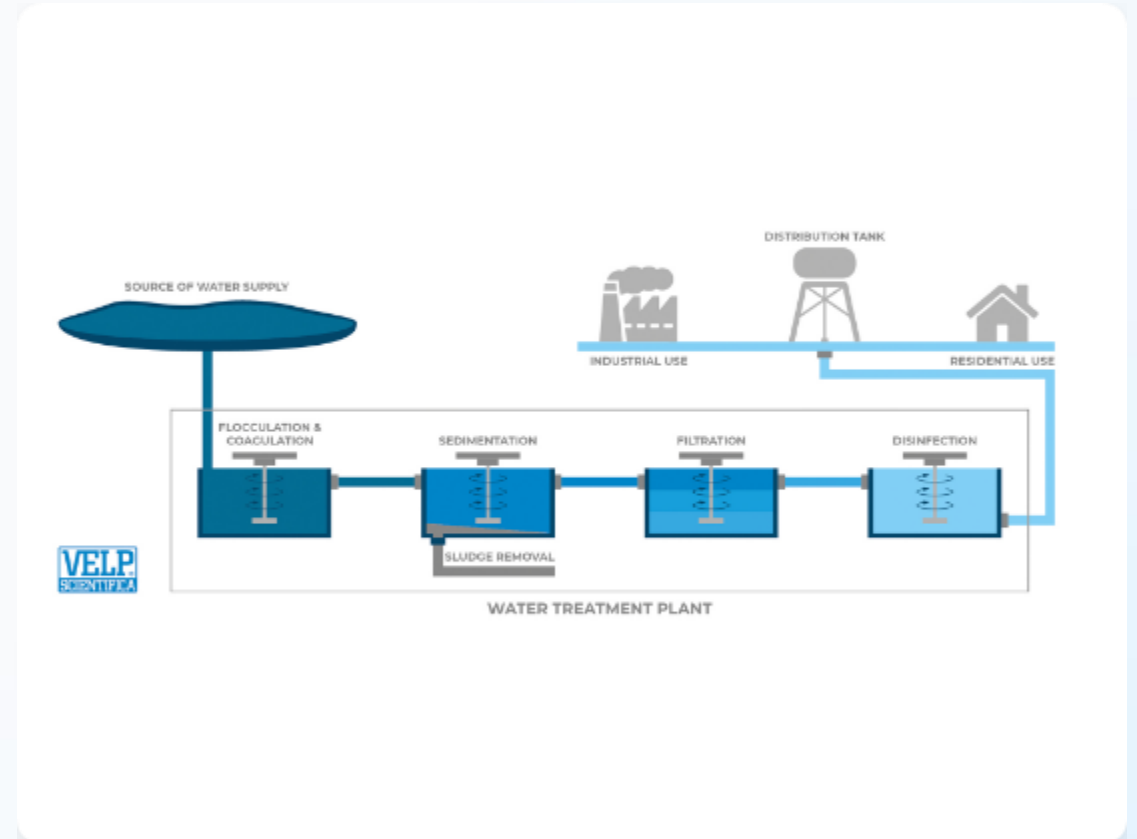
⚡ Coagulation (Fast mix):

A metal salt like alum or ferric chloride is added with intense agitation. This neutralises electrical charges that keep fine colloids apart.

❄ Flocculation (Slow mix):

A long-chain polymer (polyelectrolyte) is gently mixed in, acting as a bridge to grow destabilised particles into large, heavy, settleable "flocs".

Optimisation: The exact chemical dose is found via a "jar test". Alum operates best near pH 6–7, while ferric works over a wider pH range.



Chrome Recovery Unit

Precipitation Reaction



- 1 Collect spent chrome separately.
- 2 Raise pH with alkali (MgO or NaOH) to precipitate Cr(OH)_3 .
- 3 Settle and dewater the chrome cake.
- 4 Redissolve in H_2SO_4 to recreate reusable tanning liquor.

Why It Matters

-  Recovers a highly valuable, costly chemical resource for reuse.
-  Critically keeps toxic chromium out of the main effluent and mixed sludge.
-  Dramatically cuts the total volume of hazardous solid waste generated.
-  A mandatory, central requirement for Leather Working Group (LWG) compliance.

Primary Sedimentation

The final step of physical primary treatment.

-  **Primary Clarifier:** A large, circular settling tank equipped with slowly rotating bottom scrapers.
-  **Gravity settling:** Flocculated solids created in the previous step settle gracefully to the bottom.
-  **Detention time:** Effluent is held for 1 to 2 hours.
-  **Surface loading:** Designed for 1 to 2 m³/m²·h.

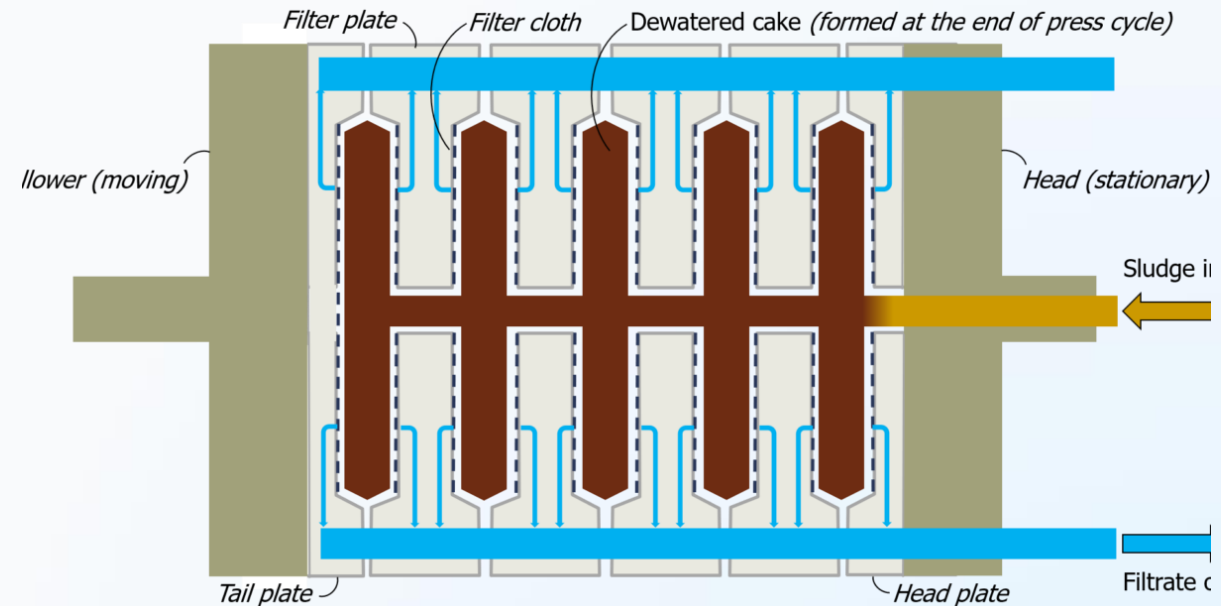
Clear, treated water overflows the top weirs to enter secondary treatment, while thickened sludge is continuously drawn from the bottom hopper.



Sludge from Primary Treatment

-  **High water content:** Raw sludge drawn from clarifiers is highly watery, containing only 2–4% dry solid mass.
-  **Dewatering:** It must be thickened and mechanically dewatered using filter presses, belt presses, or centrifuges to form a solid cake.
-  **Hazardous nature:** Chrome-bearing sludge is legally classified as hazardous. It absolutely cannot be landfilled raw.
-  **Disposal routes:** Secured (lined) landfilling, stabilisation, or dedicated incineration.

Caution: Careless incineration risks oxidizing safe trivalent chromium (Cr^{3+}) into highly toxic, carcinogenic hexavalent chromium (Cr^{6+}).



Primary Treatment at a Glance

Unit / Process	What it distinctly achieves
Coarse + fine screen	Mechanically removes 30–40% of all gross suspended solids.
Grease / grit trap	Skims floating fats and settles damaging sand before pumps.
Equalisation tank	Balances wildly varying flow, load, and pH (volume $\approx$ daily flow).
Sulphide oxidation	Chemically converts toxic S^{2-} to SO_4^{2-} ; leaves < 1 mg/L residual.
Coagulation–flocculation	Aggregates fine colloids, residual chrome, and colour into flocs.
Chrome recovery	Precipitates expensive $Cr(OH)_3$ for safe reuse, removing it from sludge.
Primary clarifier	Settles the solid flocs via gravity over a 1–2 hour detention time.
Sludge dewatering	Mechanically concentrates watery raw sludge (2–4%) into a firm cake.



Take-Home Message

Primary treatment successfully removes coarse solids, destroys sulphide gas risks, balances flow and load, corrects pH, recovers valuable chrome, and cuts massive amounts of BOD/COD.

It delivers a uniform, non-toxic feed so the biological stage can work reliably. This is exactly where environmental compliance and resource recovery are won or lost.